Bellabeat User Engagement & Sleep Tracking Analysis

Key Insight

Users consistently track activity, but only ~48% of days include sleep tracking.
Moderate users represent the largest and most valuable segment, but show inconsistent sleep tracking — making them the strongest opportunity for engagement growth and membership conversion.

Business Problem

Bellabeat aims to grow its subscription-based membership by increasing user engagement within its wellness ecosystem.

The objective of this analysis is to understand user behaviour patterns and identify opportunities to increase engagement depth, particularly in sleep and recovery tracking.

Tools Used

- SQL (data cleaning and transformation)
- Power BI (data modelling, DAX, dashboard)

Data Overview

- Dataset: Fitbit Fitness Tracker Data
- 33 users
- 910 daily activity records
- 410 sleep records (~44% of tracked days)

Approach

- Cleaned and merged activity and sleep datasets at user-day level
- Created behavioural segments:
    - Activity level (Sedentary, Moderate, Active)
    - Engagement level (High, Medium, Low)
- **Analysed relationships between:**
    - Activity behaviour
    - Sleep tracking consistency
    - User engagement levels

- Built a dashboard to highlight behavioural patterns and gaps

Key Findings

1. High Activity, Low Sleep Tracking

- Users consistently track activity
- Sleep tracking occurs on only ~44% of days

Users are engaged, but not fully utilising the platform

2. Engagement Drives Behaviour Depth

- High engagement users track both activity and sleep consistently
- Low engagement users do not track sleep at all

Engagement level directly impacts holistic tracking behaviour

3. Moderate Users Are the Largest Opportunity

- Moderate users form the largest segment
- They track activity regularly but lack consistent sleep tracking

This group is already engaged but underutilising key features

4. Activity vs Sleep Trade-Off

- **Active users:**
    - Track sleep more frequently
    - Sleep less on average
- **Sedentary users:**
    - Sleep longer
    - Track inconsistently

Behaviour is not just about activity — it reflects lifestyle patterns

Business Implications

- Users are not lacking activity engagement
- The gap lies in depth of engagement (sleep + recovery tracking)
- Increasing feature usage is more valuable than increasing user volume

Strategic Recommendations

1. Position Membership as a Holistic Wellness Product

Shift messaging from:

“Track your steps”

To:

“Understand your full wellness — activity, sleep, and recovery”

2. Target Moderate Users (Primary Segment)

Focus on users who:

- Are already active
- Track inconsistently
- Require minimal behavioural change

Tactics:

- Highlight sleep insights
- Offer partial premium previews
- Use personalised recommendations

3. Introduce Behavioural Nudges

Encourage deeper engagement through:

- Sleep tracking reminders
- Streak tracking
- Weekly summaries
- Progress-based incentives

Business Impact

Improving sleep tracking consistency can:

- Increase user engagement
- Improve perceived product value
- Drive membership conversion
- Strengthen long-term retention

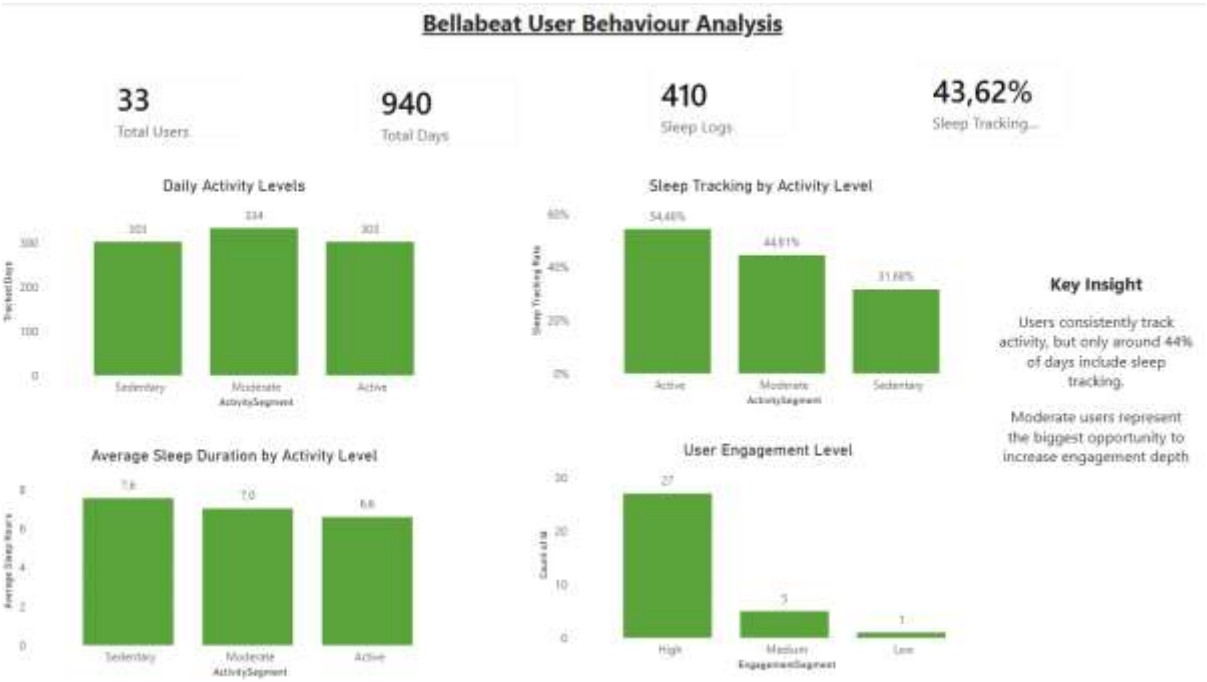
Limitations

- Small sample size (33 users)
- Limited time period (~1 month)
- No demographic or revenue data
- Fitbit users may not fully represent Bellabeat users

Conclusion

Users actively engage with fitness tracking but fail to consistently track sleep, creating a gap in holistic wellness behaviour.

Bellabeat's growth opportunity lies in increasing engagement depth rather than increasing activity usage.

Focusing on moderate users and improving sleep tracking behaviour can directly support membership growth.

Dashboard